Arunabha Majumder

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Portfolio: arunabha22.github.io/arunabha-majumder

Clira Health, Paris

Application: Mechanical Engineer, Founding Team

Dear Clira Health team,

I am applying for the Mechanical Engineer role on your founding team. Patients who need blood tests several times a week should not have to go to the hospital for each one, and building the device that fixes that is the kind of hard, useful problem I want to own.

What I have built: in my PhD at Aalborg University I designed a hybrid shoulder exoskeleton from first principles. It has a variable stiffness mechanism, with a worm gear that pretensions the spring, working in parallel with a 6 Nm motor. I optimised it with a genetic algorithm, modelled it in SolidWorks, 3D printed prototypes, had the final parts manufactured and measured its stiffness on a physical test setup. It uses more than 25% less motor energy and won the Best Paper Award at IFToMM ISRM 2026. I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM), estimated its stiffness with less than 10% error and quantified human arm impedance.

A failed experiment: my first clutch kept breaking and was hard to make with our workshop's tools. I stopped defending it and switched to a simple bolt-locking clutch that still locks at 8 angles. It taught me to design for manufacturing from day one, which matters even more for a disposable.

Before that I fabricated pneumatic artificial muscles for a soft exosuit and designed the outer PID loop of their controller (IEEE CONECCT 2022), and as a student I built an arm exoskeleton, an exo glove, a Go-Kart and a hybrid Efficycle. I also hold a Senior Diploma in Fine Arts, do graphic design in Illustrator, Photoshop and Procreate, and sketch every idea by hand before CAD.

To be honest about gaps: I have not yet designed fluidic seals or worked under ISO 13485, and I do not speak French yet, but I am willing to learn. My research is finished, so I can start as soon as possible (with time for a visa and moving), and I am ready to move to Paris. My portfolio has photos of my prototypes.

Kind regards,
Arunabha Majumder